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Dynamics Modeling of a Humanoid Robot

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Abstract. This paper presents the overall dynamics analysis for humanoid robots and proposes a method to extend the dynamics equations to check the stability conditions and compute the maximum torques required at each joint. We consider a humanoid robotics system with various base conditions. As the robot walks, the robot will switch between single and double support conditions. The conventional works on dynamical stability of the robots mostly uses simple models with invariant bases. Our proposed method models a robot with various base conditions and describes the dynamics equations of humanoid robots in three dimensional space. Using this method, a straight walking was simulated as a sample gait during which joint torques was achieved and the stability condition was tested.

1 Introduction

Large number of degrees-of-freedom in humanoid robots limits most constructed prototypes to use servomotors, so that each motor is controlled independently from the others. This results in maximum torques and response times of the motors become their most significant property. Therefore, one of the most important problems during the design process is to determine the maximum torque and angular velocity of each joint, which is the result of appropriate dynamical model and gait generation. Meanwhile, another important issue in controlling a humanoid robot is the equilibrium conditions.

There are two ways to investigate the dynamical equilibrium conditions of a humanoid robot. The first method, which was under research in many laboratories, is to find the ZMP (zero moment point) of a robot and check whether it is in an allowable margin, see [1]. The position of the ZMP can be calculated from the linear and angular velocity and acceleration of links of the robot or from the interaction forces of robot's feet with the ground. The second method and more realistic one is dynamical modeling of the robot. Having the joint angles for robots with servos (which is computable from a desired robot gait), and assuming an ideal condition (in which motors can provide the required torques), this method results in equilibrium condition during walking. Considering these two results, we will try to introduce a three dimensional dynamical model.

There are many papers about humanoid robots that introduce a simple, (mostly two-dimensional) model to provide a proper platform for their control theory. The simplest 5-link model has been introduced in many publications. Ching-Long has dealt with the inverted pendulum modeling of this simple model [2]. Also Tsu-Tian, Jeng and Gruver have treated trajectory planning and steady walking control of this model [3]. This is a sagittal projection of walking robot in which the frontal degrees-of-freedom does not have an effective influence on lateral stability of the robot. The main goal of this paper is to model a humanoid robot in three dimensions and check for stability as well as and torque determination of required joint torques.

2 Description of High Level Model

In this paper, the base of a humanoid robot is defined as a part of the robot in force interaction with the ground. One of the most common approaches in robot dynamics is Iterative Newton-Euler method. In our humanoid robot dynamic formulation combination of prismatic and revolute joints are considered and the ground is assumed as the link zero and other links are numbered successively. As it is understood from the brief description of this method, a link zero is considered as a stationary link or a link with a known velocity and acceleration.

A new method is proposed for dynamics of robot with various base situation by using one of the probable bases as a reference base, and continuing the iterative method by assuming a known velocity and acceleration for it and then extracting the base dynamics as a three dimensional body. The important issue in this method is how to implement proper connection between the relations for the robot body and the reference base so that a solvable equation set is reached.

Its bases perform the interaction of robot with environment. In general, we have to use a contact model for the robot interactions with the ground, where the contact forces would be functions of kinematical parameters of the base. The relations and dependencies between different parts of the robot are illustrated below.

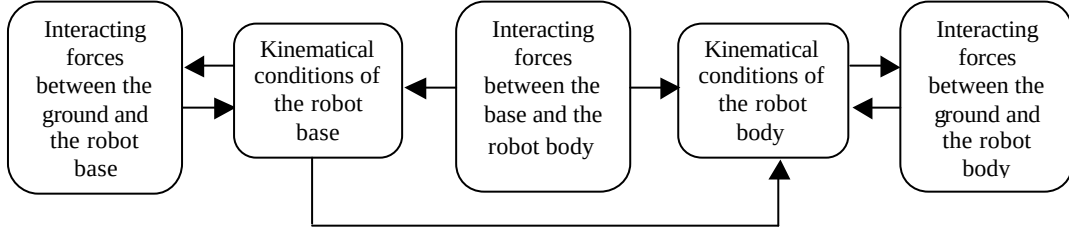


Fig. 1. Dependencies among robot components.

Generally, the aforementioned dependences can be classified as followed:

- Relations among kinematical parameters, forces and torques of every part (body and reference base), which illustrates their dynamics.
- Dependence of kinematical situation of the body to the base kinematical conditions, which is inevitable regarding the chosen structure.
- Relations involving forces and torques exerted on common joints of the body and the base.

To model base contact, four corner points of the robot foot is considered as contact points, where spring-damper model was applied for normal forces, and a friction model for tangential forces. Using Young's modulus coefficients of restitution, the normal force can be formulated as:

$$f_v(z) = K z + z' z' \quad (2.1)$$

Where K is the modulus of elasticity, z is the structural damping coefficient of the contact material and z denotes the penetration depth of the contact point.

There are some friction models, which mostly model the friction in dry sliding contacts between two flat surfaces as elastic and plastic deformation forces of tiny asperities in the contact region. One of them is the Dahl friction model [6], which was developed for the purpose of simulating frictional control systems. To obtain a time domain, Dahl observed that

$$\frac{dF}{dt} = s \left(1 - \frac{F}{F_c} \operatorname{sgn}(v)\right)^a v \quad (2.2)$$

The model is a generalization of conventional Coulomb friction F_c . There exist several recent extensions of this model, see [7, 10]. The major problem for using this method is the differential form of friction force, which does not let us reset F in differential equations during the period that the base is raising from the ground. The LuGre model is a dynamic friction model presented in [8]. Extensive analysis of the model and its application can be found in [9]. The model is related to the bristle interpretation of friction as in [7]. Friction is modeled as the average deflection force of elastic springs. When a tangential force is applied, the bristles will deflect like springs. If the deflection is sufficiently large the bristles start to slip. The velocity determines the average bristle deflection for a steady state motion. It is smaller at low velocities, which implies that the steady state deflection decreases while the velocity increases. This models the phenomenon that the surfaces are pushed apart by the lubricant, and models the Stribeck effect. The model also includes rate dependent friction phenomena such as varying breakaway force and frictional lag. The model equations have the form:

$$\begin{aligned} \frac{dz}{dt} &= v - \frac{s_0 |v|}{g(v)} z \\ F &= s_0 z + s_1(v) \frac{dz}{dt} + f(v) \end{aligned} \quad (2.3)$$

Where z denotes the average bristle deflection, the parameter s_0 is the stiffness of the bristles, and $s_1(v)$ the damping. The model behaves like a spring for small displacements. For constant velocity, the steady state friction force can be obtained from:

$$F = g(v) \operatorname{sgn}(v) + f(v) \quad (2.4)$$

The function $g(v)$ models the Stribeck effect (increasing friction coefficient versus velocity) [13], and $f(v)$ is the viscous friction. A reasonable choice of $g(v)$ which gives a good approximation of the Stribeck effect is:

$$g(v) = a_0 + a_1 e^{-\left(\frac{v}{v_0}\right)^2} \quad (2.5)$$

The sum $a_0 + a_1$ then corresponds to the static force and a_0 corresponds to Coulomb friction force.

The following special case of the model given by equation (2.6), which has linear viscous friction and constant, s_1 is called the standard parameterization.

$$\begin{aligned}\frac{dz}{dt} &= v - \frac{s_0 |v|}{g(v)} z, \\ g(v) &= a_0 + a_1 e^{-(v/v_0)^2} \\ F &= s_0 z + s_1 z' + a_2 v\end{aligned}\tag{2.6}$$

It is useful to let the damping s_1 decrease with increasing velocity, e.g.

$$s_1(v) = s_1 e^{-(v/v_d)^2}\tag{2.7}$$

Physically, this is motivated by the change of the damping characteristics as velocity increases, due to more lubricant being forced into the interface. Another reason for using (2.7) is that it gives a model, which is dissipative [9]. Applying this method and equation (2.1) interaction force equations will be:

$$\begin{aligned}f &= -u(z) \cdot \begin{bmatrix} s_0 z_x + s_1 z'_x + a_2 v_x \\ s_0 z_y + s_1 z'_y + a_2 v_y \\ K z + z' \end{bmatrix} \\ \frac{dz_x}{dt} &= v_x - \frac{s_0 |v_x|}{g(v_x)} z_x \\ \frac{dz_y}{dt} &= v_y - \frac{s_0 |v_y|}{g(v_y)} z_y \\ g(v) &= a_0 + a_1 e^{-(v/v_0)^2}\end{aligned}\tag{2.8}$$

3 Robot Body Dynamics

The robot body consists of successive links. The first link denotes the reference frame. To simplify the equations, the robot is considered as a biped model. It is feasible to generalize these relations throughout the whole robot body to take into account the effect of upper body. The frames are settled by starting from one of the robot feet as the reference base and then continuing to its shank, thigh and hip links, and afterwards moving inversely in the other leg to reach its foot.

The parametric kinematical condition of robot is required for dynamical analysis. The Denavit-Hartenberg method [4] will be used, so that a particular frame is affixed to each link in order to define

the relative position of the frames. Four parameters are defined for each intermediate link: joint angle variable q and three constants a , d , a . The frames attached to the links of robot are shown in Fig. 1 and Table 1 involves the parameters used in the proposed method. The transform matrix ${}^{i-1}_iT$ which defines frame i relative to frame $i-1$ is:

$${}^{i-1}_iT = \begin{bmatrix} r_{11} & r_{12} & r_{13} & a \\ r_{21} & r_{22} & r_{23} & b \\ r_{31} & r_{32} & r_{33} & c \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \cos q_i & -\sin q_i & 0 & a_{i-1} \\ \sin q_i \cos a_{i-1} & \cos q_i \cos a_{i-1} & -\sin a_{i-1} & -\sin a_{i-1} d_i \\ \sin q_i \sin a_{i-1} & \cos q_i \sin a_{i-1} & \cos a_{i-1} & \cos a_{i-1} d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (3.1)$$

This consists of a 3 x 3 rotation and 3 x 1 displacement matrices. Another frame will be affixed to the end of the last link in order to determine the location of the other foot. Having spatial transformations of the robot links relative to each other, robot dynamics would be acquired.

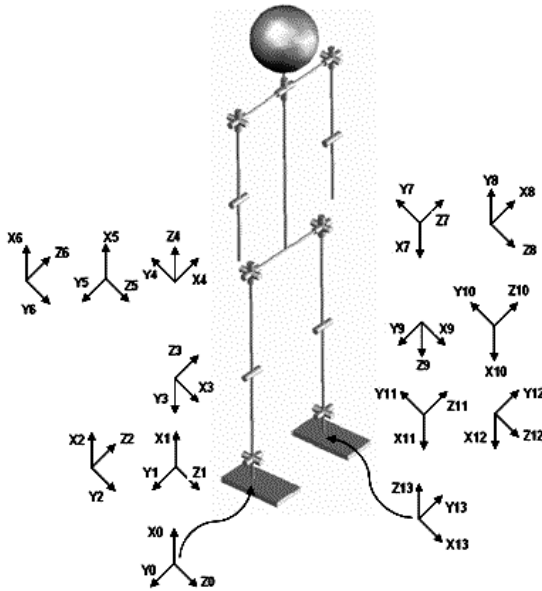


Fig. 2. The robot frames allocation

i	a_{i-1}	a_{i-1}	d_i	q_i
1	Malleolus	0	0	q_1
2	p/2	0	0	q_2
3	0	Shank	0	q_3
4	p/2	0	Thigh	q_4
5	p/2	0	0	q_5
6	p/2	0	0	q_6
7	0	0	Hip	q_7
8	p/2	0	0	q_8
9	-p/2	0	Thigh	q_9
10	p/2	0	0	q_{10}
11	0	Shank	0	q_{11}
12	p/2	0	0	q_{12}
13	Malleolus	0	0	0

Table 1. Frame affixing Parameters

The Newton-Euler iterative method [5], which consists of two outward and inward series of iterations with the following relations, is used [11].

Outward Equations($0 \rightarrow n-1$)

$$\begin{aligned}
{}^{i+1}\mathbf{w}_{i+1} &= {}^iR^{i+1}\mathbf{w}_i + \dot{q}_{i+1} {}^{i+1}\hat{\mathbf{Z}}_{i+1} \\
{}^{i+1}\dot{\mathbf{w}}_{i+1} &= {}^iR^{i+1}\dot{\mathbf{w}}_i + {}^iR^{i+1}\mathbf{w}_i \times \dot{q}_{i+1} {}^{i+1}\hat{\mathbf{Z}}_{i+1} + \ddot{q}_{i+1} {}^{i+1}\hat{\mathbf{Z}}_{i+1} \\
{}^{i+1}\dot{\mathbf{v}}_{i+1} &= {}^iR^{i+1}(\dot{\mathbf{w}}_i \times {}^iP_{i+1} + {}^i\mathbf{w}_i \times ({}^i\mathbf{w}_i \times {}^iP_{i+1}) + {}^i\dot{\mathbf{v}}_i) \\
{}^{i+1}\dot{\mathbf{v}}_{C_{i+1}} &= {}^{i+1}\dot{\mathbf{w}}_{i+1} \times {}^{i+1}P_{C_{i+1}} + {}^{i+1}\mathbf{w}_{i+1} \times ({}^{i+1}\mathbf{w}_{i+1} \times {}^{i+1}P_{C_{i+1}}) + {}^{i+1}\dot{\mathbf{v}}_{i+1} \\
{}^{i+1}F_{i+1} &= m_{i+1} {}^{i+1}\dot{\mathbf{v}}_{C_{i+1}} \\
{}^{i+1}N_{i+1} &= {}^{C_{i+1}}I_{i+1} {}^{i+1}\dot{\mathbf{w}}_{i+1} + {}^{i+1}\mathbf{w}_{i+1} \times {}^{C_{i+1}}I_{i+1} {}^{i+1}\mathbf{w}_{i+1}
\end{aligned} \tag{3.2.a}$$

Inward Equations($n \rightarrow 1$)

$$\begin{aligned}
{}^if_i &= {}^{i+1}R^{i+1}f_{i+1} + {}^iF_i \\
{}^in_i &= {}^iN_{i+1} + {}^{i+1}R^{i+1}n_{i+1} + {}^iP_{C_i} \times {}^iF_i + {}^iP_{i+1} \times {}^{i+1}R^{i+1}f_{i+1} \\
t_i &= {}^in_i {}^i\hat{\mathbf{Z}}_i
\end{aligned} \tag{3.2.b}$$

The special symbols used in the above equations are described in appendix.

During outward iterations in computing velocities and accelerations, kinematical parameters of the base are assumed known. In the next section (Reference Base Dynamics), these parameters will be discussed and calculated. Also in inward iterations while computing force and torque of the joints, using the contact model, the contact forces of the non-reference foot, should be defined. It should be noted that the needed position and velocities for calculating these forces are already in hand from inward equations. Equations of this section can be applied to obtain position and velocity of the other foot (inward equations) and to calculate joint torques (outward equations). To include gravity forces for links, gravity acceleration is added to the acceleration of reference base with negative sign along z-axis.

4 Reference Base Dynamics

In this section the main concentration is to determine the position and orientation of the reference base (right foot) relative to the reference frame. Representing the link zero, this frame is affixed to the ground with x-axis along the walking direction and z-axis toward the shank and all parameters in this

section are described in this frame. The position of center of mass of the right foot is defined by $\vec{d}$ and $\vec{R}$ denotes the rotation matrix of the base frame. In addition 1a , 1b and 1g are roll, pitch and yaw angles for this frame. The link one (reference base) is represented by the frame affixed to its center of mass. The initial conditions specify the initial orientation of this link relative to reference frame (link zero). To calculate R, angular velocity matrix of frame one (${}^0_1\Omega$) and subsequently (0w) is required.

$${}^0_1\Omega = \begin{pmatrix} 0 & -{}^0\dot{a} & {}^0\dot{b} \\ {}^0\dot{a} & 0 & -{}^0\dot{g} \\ -{}^0\dot{b} & {}^0\dot{g} & 0 \end{pmatrix} \quad (4.1)$$

$${}^0w = R {}^1w, \quad {}^1w = \begin{pmatrix} {}^1\dot{g} \\ {}^1\dot{b} \\ {}^1\dot{a} \end{pmatrix}$$

Where 1w is calculated by integrating the torque equation of the right foot [12]:

$$\sum M = \sum (\vec{r}_i \times R \vec{f}_{hi} + \vec{r}_i \times R \vec{f}_{vi}) - \vec{n}_1 - \vec{l}_o \times \vec{f}_1 = I_{G_b} {}^1\dot{w} + \begin{pmatrix} 0 & -{}^1\dot{a} & {}^1\dot{b} \\ {}^1\dot{a} & 0 & -{}^1\dot{g} \\ -{}^1\dot{b} & {}^1\dot{g} & 0 \end{pmatrix} I_{G_b} {}^1w \quad (4.2)$$

Where r_i is the position vector of the contact point i described relative to frame one, I_{G_b} is the inertia tensor of the foot relative to its own frame, $\vec{n}_1$ and $\vec{f}_1$ are respectively the torque and the force exerted on link two by base link and l_o denotes position of joint one in frame one. The exerted forces on the right foot are divided into normal forces f_{vi} and frictional forces f_{hi} . Having calculated ${}^0_1\Omega$, it will be possible to determine matrix R with integrating the following equation

$$\dot{R}_{(t)} = {}^0_1\Omega R \quad (4.3)$$

In order to calculate $\vec{d}$ the force equation of the base is needed [12]:

$$\sum F = \sum (\vec{f}_{hi} + \vec{f}_{vi}) + m_b \vec{g} - R^{-1} \vec{f}_1 = m_b \ddot{\vec{d}} \quad (4.4)$$

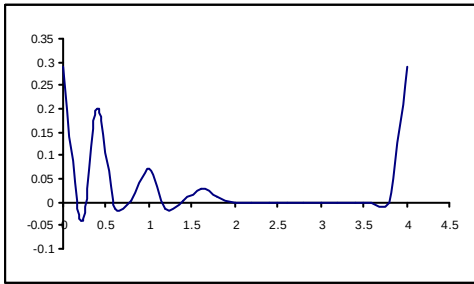
Where mass of foot is denoted by m_b . To calculate $\vec{f}_{vi}$ and $\vec{f}_{hi}$ in the force and torque equations (4.2, 4.4), the position and the velocity of the contact points are required:

$$\begin{aligned}\vec{l}_i &= \vec{d} + R^{-1} \vec{r}_i \\ \dot{\vec{l}}_i &= \dot{\vec{d}} + R^{-1} {}^1_0\Omega \vec{r}_i\end{aligned}\tag{4.5}$$

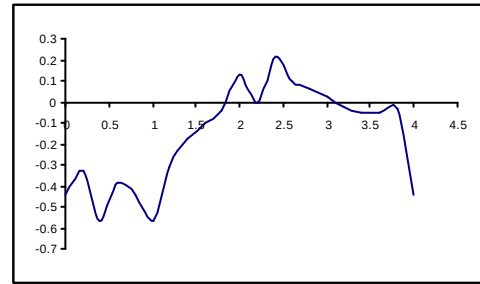
Having the position and velocity of contact points relative to the reference frame and using the equations (2.8), the normal and frictional forces relative to frame one can be calculated. Generally, the number of contact points is arbitrary and depends on simulation accuracy level. In our simulation, to model foot-ground contact, four-corner contact points (the minimum number that should be considered) are used.

5 Simulation and Results

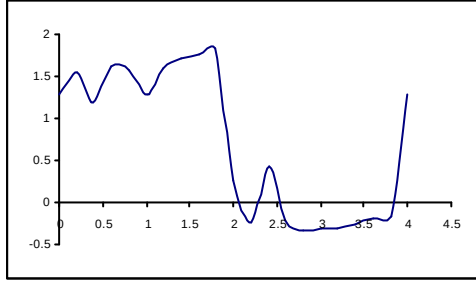
Using the proposed method, simulation of stable walking was conducted by considering mass and inertia tensors proportional to human body. The involved gait was extracted from straight human walking trajectory. Each robot leg consists of three sets of joints - ankle, knee and hip joints. The trajectories of hip and ankle joints were generated independently and knee joint trajectory was determined regarding the lengths of shank and thigh links. Inverse kinematics was applied to joint trajectories and angle of each joint was calculated during the gait. Employing resulted joint angle trajectories, dynamical differential equations were solved to calculate the corresponding joint torques, and to check the robot stability. The following charts show the amount of needed joint torques during the robot walking.



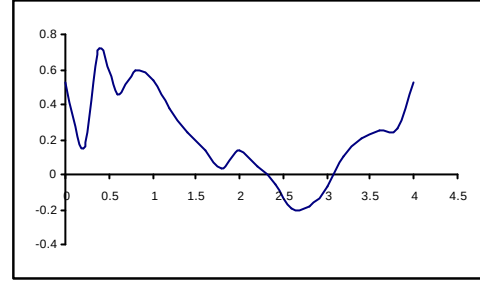
(a)



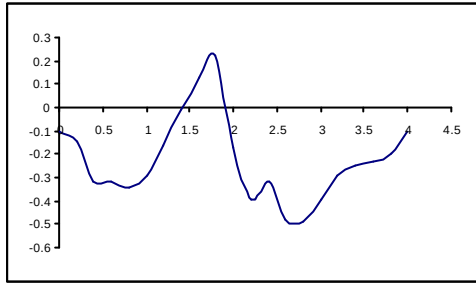
(b)



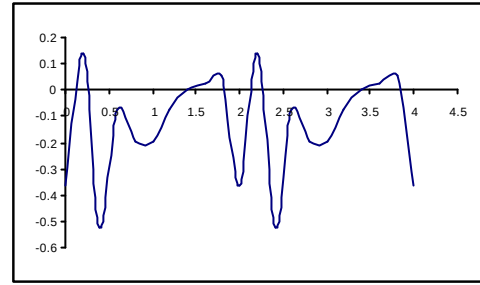
(c)



(d)



(e)



(f)

Fig. 3. Joint torques versus time during one period of the robot walking: (a) Ankle X, (b) Ankle Y, (c) Knee, (d) Hip Z, (e) Hip X, (f) Hip Y

6 Conclusions and Future Work

In order to obtain a robust motion control for humanoid robots, it is necessary to have an accurate dynamical model based on both experiments and theories. The dynamical model proposed in this paper has the great advantage of being more realistic over conventional two-dimensional models. The present model is simple and it captures many but not all aspects relevant to humanoid robot modeling.

However several improvements remain for the future. It is necessary to introduce a method for identifying more accurate simulation parameters and test the model for other trajectories like slope walking or climbing up or down the staircases. The current work is new and very important in robot dynamics, and the future work will be on extending the algorithms in these directions, as well as conducting practical experiments on the Sharif Humanoid Robot.

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Appendix

The notations used in equations (3.2.a and 3.2.b) proposed by Denavit and Hartenberg [4] are as follows. A detailed description of Denavit-Hartenberg method applied on Newton-Euler dynamics equations can be found in [11].

n : Number of robot links

${}^{i+1}\mathbf{w}_{i+1}$: Absolute angular velocity of frame i+1 relative to its own frame

${}^{i+1}_i\mathbf{R}$: Transverse matrix of rotation consisting of r_{11} to r_{33} in ${}^{i-1}_i\mathbf{T}$

$\dot{\mathbf{q}}_{i+1} {}^{i+1}\hat{\mathbf{Z}}_{i+1}$: Angular velocity of joint i+1

${}^{i+1}\dot{\mathbf{w}}_{i+1}$: Absolute angular acceleration of frame i+1 relative to its own frame

$\ddot{\mathbf{q}}_{i+1} {}^{i+1}\hat{\mathbf{Z}}_{i+1}$: Angular acceleration of joint i+1

${}^{i+1}\dot{\mathbf{v}}_{i+1}$: Absolute linear acceleration of frame i+1 relative to its own frame

${}^i\mathbf{P}_{i+1}$: Position of origin of frame i+1 relative to frame i

${}^{i+1}\dot{\mathbf{v}}_{C_{i+1}}$: Absolute linear acceleration of center of mass of link i+1 relative to its own frame

${}^{i+1}\mathbf{P}_{C_{i+1}}$: Position of center of mass link i+1 relative to its own frame

${}^{i+1}\mathbf{F}_{i+1}$: Inertial force exerted on link i+1

m_{i+1} : Mass of link i+1

${}^{i+1}\mathbf{N}_{i+1}$: Inertial torque exerted on link i+1

${}^{C_{i+1}}\mathbf{I}_{i+1}$: Inertial tensor of link i+1

${}^i\mathbf{f}$: Force exerted on link I by link i-1

${}^i\mathbf{n}_i$: Torque exerted on link I by link i-1

$\mathbf{t}_i$: Joint torque of link i